



Exploring the Relationship Between Trader Performance and Market Sentiment

Internship Assessment Report

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Colab Notebook [View analysis notebook](#)

1. Objective

The objective of this analysis is to investigate the relationship between Bitcoin market sentiment and trader performance, using historical trading data from Hyperliquid together with the Bitcoin Fear & Greed Index. The analysis focuses on three goals:

- Explore the relationship between trader performance and market sentiment.
- Uncover hidden patterns in trading behaviour across different market conditions.
- Deliver insights that can support smarter, adaptive trading strategies.

2. Dataset Description

Two datasets were used and merged on the trading date, resulting in **83,798 matched trade records**.

Bitcoin Fear & Greed Index	Hyperliquid Historical Trading Data
Daily market sentiment classification — Extreme Fear, Fear, Neutral, Greed, Extreme Greed — plus a numerical sentiment score (0–100).	83,805 historical trades, including trader account, coin, execution price, trade size, buy/sell side, closed P&L, fees, and timestamp.

3. Methodology

- Loaded and inspected both datasets; removed incomplete records.
- Standardized timestamp formats and merged the datasets on trading date.
- Performed exploratory data analysis and compared trader performance across sentiment categories.
- Conducted statistical validation using the Shapiro-Wilk and Kruskal-Wallis tests.
- Generated visualizations to support the findings.

4. Relationship Between Trader Performance and Market Sentiment

The merged dataset was grouped by market sentiment category:

Sentiment	Trades	Avg Closed PnL	Win Rate
Extreme Fear	7,172	87.44	38.02%
Fear	30,401	82.17	45.79%
Greed	19,586	79.75	40.29%
Neutral	15,928	71.85	46.53%
Extreme Greed	10,711	155.32	53.86%

Observations

- Trader profitability varies across market sentiment categories.
- **Extreme Greed** recorded both the highest average Closed PnL and the highest win rate.
- Neutral market conditions produced the lowest average profitability.
- These findings indicate that trader performance is influenced by the prevailing market sentiment.

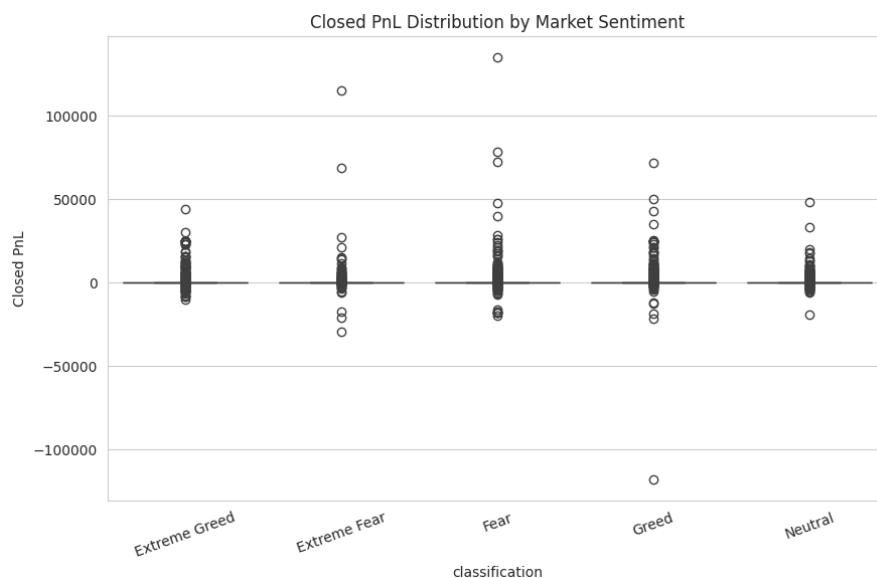


Figure 1. Closed PnL distribution across market sentiment categories, showing heavy-tailed outliers in every regime.

5. Hidden Patterns

Pattern 1 — Sentiment influences groups, not individual trades

The Pearson correlation between the Fear & Greed score and Closed PnL is only **0.015** — an extremely weak relationship. This indicates that market sentiment does not explain the outcome of any single trade; rather, it shapes the overall trading environment.

Pattern 2 — Profit distribution is highly skewed

Every sentiment category shows a large number of outliers. Median Closed PnL stays close to zero, while a small number of highly profitable trades drive most of the average profitability — returns remain volatile regardless of sentiment.

Pattern 3 — Buy/Sell performance shifts with sentiment

SELL trades outperform BUY trades during Extreme Greed, while BUY trades outperform SELL trades during Fear — the effectiveness of trade direction depends on the market regime.

Pattern 4 — Trader performance differs across conditions

Trader-level analysis shows that different traders perform best under different sentiment categories, suggesting experienced traders adapt strategy rather than following one fixed approach.

Pattern 5 — Cryptocurrency performance is sentiment-dependent

The coin-vs-sentiment heatmap shows that different assets respond differently under the same sentiment — some consistently outperform during Fear, others during Greed or Extreme Greed — supporting asset-specific trading strategies.

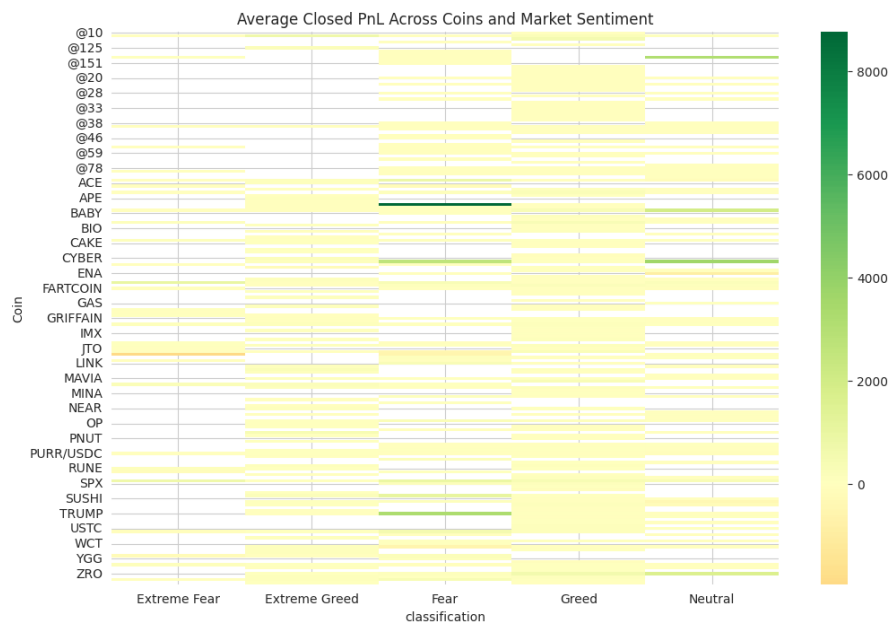


Figure 2. Average Closed PnL across coins and market sentiment categories.

6. Statistical Validation

Test	Purpose	Result	Conclusion
Shapiro-Wilk	Test normality of Closed PnL	Stat = 0.0285 p = 3.61×10^{-95}	Closed PnL is not normally distributed
Kruskal-Wallis	Compare profitability across sentiment groups	H = 836.45 p = 9.77×10^{-180}	Profitability differs significantly by sentiment

Both p-values fall far below the 0.05 significance threshold. This confirms that Closed PnL is non-normally distributed, and that the observed differences in trader profitability across sentiment categories are statistically significant and unlikely to have occurred by chance.

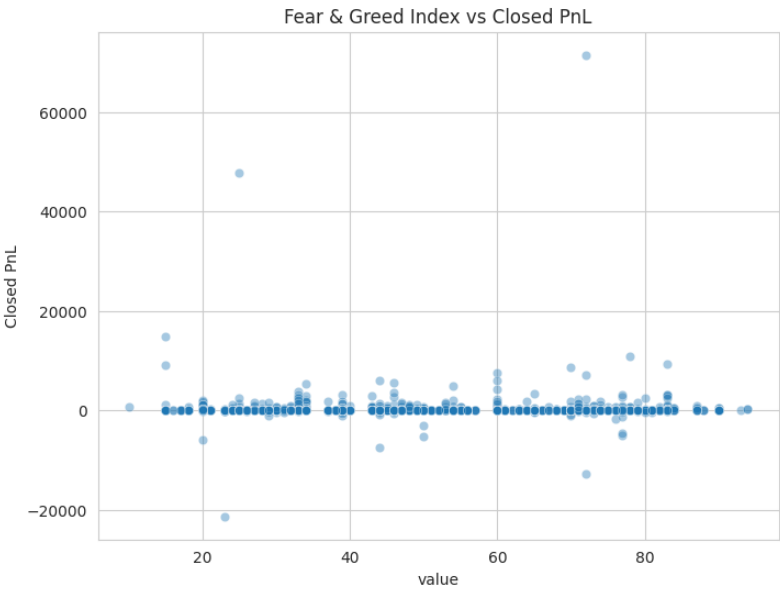


Figure 3. Fear & Greed Index score vs Closed PnL — near-zero linear relationship ($r = 0.015$).

7. Trading Insights

- 1. Use sentiment as context, not a signal**
The weak correlation (0.015) means sentiment should not be used as a standalone trading signal — it should complement technical indicators and risk management.
- 2. Extreme Greed historically produced the strongest performance**
Both the highest average Closed PnL and win rate occurred during Extreme Greed, though traders should stay cautious given the elevated profit volatility in this regime.
- 3. Strategy should adapt to sentiment**
BUY and SELL trades perform differently depending on prevailing sentiment, so adaptive strategies are likely to outperform static approaches.
- 4. Different assets require different strategies**
Cryptocurrencies do not respond uniformly to sentiment shifts, favouring asset-specific strategies over a single universal rule.
- 5. Let statistical evidence guide decisions**
The Kruskal-Wallis result confirms sentiment-driven differences in profitability are significant, supporting its use as one input among several in a trading model.

8. Conclusion

This study explored the relationship between Bitcoin market sentiment and trader performance by integrating the Fear & Greed Index with Hyperliquid historical trading data. The analysis shows a statistically significant relationship between market sentiment and trader profitability; however, the extremely weak linear correlation between sentiment scores and individual trade outcomes indicates that sentiment alone cannot predict profitability.

Market sentiment is best treated as a contextual indicator that supports decision-making alongside technical analysis, trader behaviour, asset selection, and disciplined risk management. Overall, adaptive strategies that combine sentiment with quantitative indicators are more likely to deliver consistent performance than strategies relying on sentiment alone.